Quotation No : Q23000080

Date : 21-Feb-2023

SUNIL SPONGE PVT. LTD.
SILTARA INDUSTRIAL AREA RAIPUR
RAIPUR
CHATTISGARH

Kind Attention : Mr./Ms. SUNIL MEGHANI

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

Absstem Industrial Solutions Pvt. Ltd.
Ground Floor, Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122052
Contact: 1800 547 0506 | indsales@absstem.com
CIN: U29256HR2022PTC103387

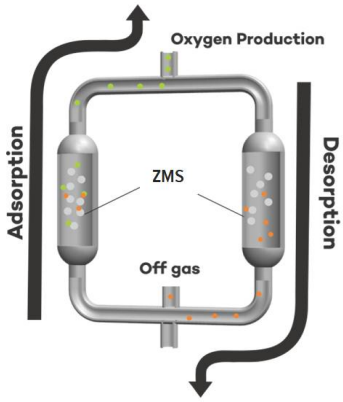
POWERED BY ABSSTEM INDUSTRIAL

Industrial

PSA Oxygen Generator



PSA Technology



PSA Process explaining oxygen production

Industrial Oxygen Generator mainly consists of two adsorbers filled with zeolite molecular sieves. Pressurized air is passed through one of these adsorbers, zeolite having a property to attract nitrogen, adsorbs Nitrogen from air and oxygen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Oxygen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of oxygen produced is monitored by an Oxygen analyser.

Absstem Industrial solutions Pvt. Ltd.
Corporate Address: *Plot No. 591, Phase- IV, Sector 8, IMT Manesar, Gurugram, Haryana-122052*
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Key Features

- Shipped ready for installation
- 5 minutes Startup time with uninterrupted stoppages.
- Quick & Easy installation
- Low operating costs compared to gas cylinders/liquid oxygen
- Reduced Maintenance costs
- Embedded Safety Interlocks to cater any abnormality.
- Space-saving compact design.
- Eliminates all safety hazards which are there with gas cylinder or liquid oxygen.
- 10 years warranty on molecular sieves.

Description

a. Product	Oxygen Gas Generator
b. Model	**IDOS**
c. Technology	PSA (Pressure Swing Adsorption)

Specifications

a.	Oxygen Flow Rate	As per Performance Sheet
b.	Oxygen Purity	As per Performance Sheet
c.	Oxygen Delivery Pressure	5.5 Barg (Higher Pressure Options Available)
d.	Ambient conditions	
	• Temperature	0 to 50 Deg. Celsius
	• Relative Humidity	up to 80 % (Max.)
	• Altitude	up to 1000 m above Sea level

Scheme Offered

This scheme will comprise of

- Air package
  (Air compressor, Air dryer, Air Receiver, Water Separator)
- Industrial Oxygen generator with oil removal filtration system
- Storage Tank
- Oxygen Analyzer
- Alarm panel

Unique Features of Absstem

a. All fabricated items are cleaned and degreased as per GEP standards.
b. All Instruments used are compatible with oxygen gas.
c. Guaranteed 10+ years of trouble free and reliable plant operation.
d. We provide the purest **Oxygen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

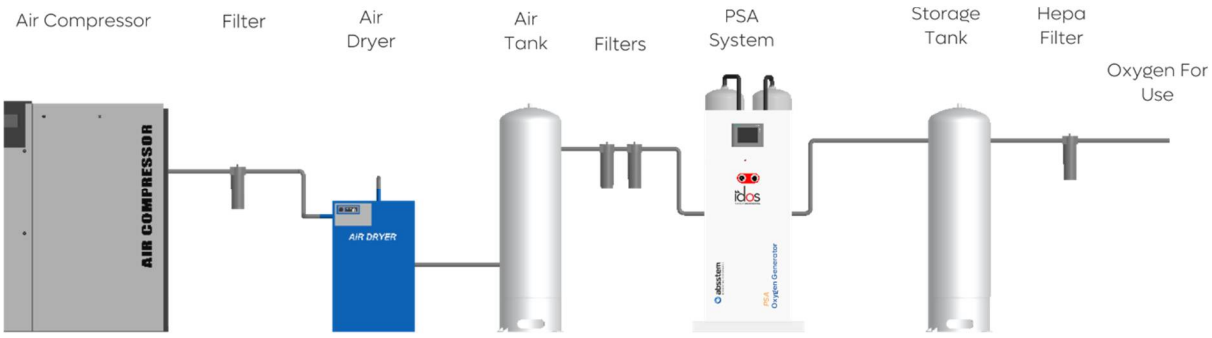
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Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 x 7 across 365 days in an year
Easy integration within existing installations
Your own independent supply source of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards related to handling high-pressure cylinders

Equipment Description

The Industrial Oxygen Generator consists of



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **oxygen** from the air.
Refrigerated Air Dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Receiver	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Oxygen** from Air.
Storage Tank	Industrial Grade **Oxygen** produced from IDOS is stored in this tank to supply a continuous flow of **Oxygen**.
Online Oxygen Analyzer	Online Oxygen analyzer for continuous monitoring of Oxygen purity level.

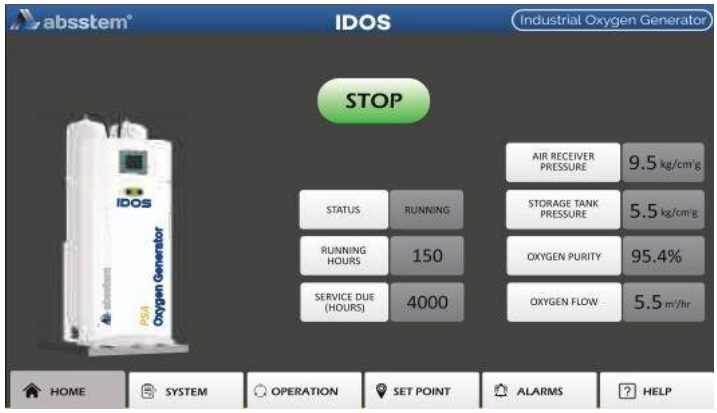
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HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial oxygen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation



Additional upgrades

- Digital flow meter with totaliser
- High-pressure oxygen tank
- Oxygen booster for high pressure requirements
- Cylinder manifold and alarm management system
- Remote monitoring

Plant Operation

- Unit operates 24-hrs a day, 7-days a week throughout the year reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches a preset limit. This helps in optimised Air Compressor operation and reduced power consumption.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial oxygen generators.

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Itemised Supply description

No.	Item description	Make	Qty
a	Air Compressor	Kaeser/IR/As per Demand	Client scope/ Quoted Separately
b	Refrigeration Dryer with pre filter	Kaeser/IR/As per Demand	Client scope/ Part of Compressor Package.
c	Air Tank	ABSSTEM	Client scope/ Part of Compressor Package.
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (IDOS)	ABSSTEM	1 no.
f	Electrical Power Panel with PLC and cabling	ABSSTEM & MITSUBISHI	1 no.
f	Digital Oxygen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	Quoted Separately.

Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Oxygen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from Rittal are IP-55 rated making the electrical enclosure internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Maintenance Contract (AMC / CMC)

Our in-house service support packages Absstem Shield & Absstem Shield premium ensures 24*7 on site support for all scheduled and need based service requirements. Per the schedule mentioned in the operational manual, we suggest the user to change the filters by following a simple protocol.

Our technical experts are always available to guide your maintenance team for carrying out minor troubleshooting and basic maintenance activities via remote support program.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. IDOS is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

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Product Description

Product	Description	Make	Qty
IDOS Air Package	IDOS 15 CP Oxygen		1

Caption	Specification
Air Compressor KW	18 KW
Air Compressor Make	Atlas Copco/ IR/ Power Efficient Brands
Air Dryer	Included
Inter Connecting Cable	Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	115 CFM

Product	Description	Make	Qty
IDOS Gas Storage	IDOS 15 ST Oxygen		1

Caption	Specification
Water Volume	3000 Litres
Gas Stored	Oxygen
Gas Stored Volume	16.5 M3
Working Pressure	5.5 Barg
Hydrotest Pressure	9 Barg

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Product	Description	Make	Qty
IDOS Oxygen Generator	IDOS 15 B Oxygen		1

Caption	Specification
Gas Generated	Oxygen
Flow	15 Nm3/Hr
Purity	95 Percent
Compressor	In Your Scope / Quoted Separately
Dryer	In Your Scope / Part of Compressor Package
Air Vessel	In Your Scope / Part of Compressor Package
Dry Air Required	115 CFM
Power Required	230V 50 Hz Single Phase
Module Option	PSA Plant Only

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	IDOS 15 CP	IDOS 15 CP	IDOS Air Package	8414	1	10,70,400.00	18.00	10,70,400.00
2	IDOS 15 ST	IDOS 15 ST	IDOS Gas Storage	842139	1	2,52,000.00	18.00	2,52,000.00
3	IDOS 15 B	IDOS 15 B	IDOS Oxygen Generator	842139	1	24,99,600.00	18.00	24,99,600.00
							Sub Total	38,22,000.00
							GST(%) @18.00	0.00
							Grand Total ()	45,09,960.00

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

www.absstem.com

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idos

POWERED BY **ABSSTEM INDUSTRIAL**



REINVENTING PERFORMANCE AND RELIABILITY

Industrial on-site PSA Oxygen Generator

Rise above the usual industrial challenges

of high-pressure cylinders and liquid oxygen tanks.

It's time to end the 6-major issues plaguing your industry such as:

Unprecedented Price Hikes

Bottled and liquefied gases are subject to fluctuating prices.

Demand & Supply Gap

The highly dynamic nature of the market, with multiple vendors having different supply capabilities, and distribution at mission-critical moments, can cause a large gap in supply when demand rises.

Remote Location Access

The more remote the location, the greater the transportation and logistical costs.

Evaporation Losses

5-10% of evaporation and non-usage losses are observed.

Storage & Handling Safety Concerns

Oxygen cylinders are heavy and need to be handled by a well-trained team.

Negligence and no Supply

Running out of oxygen is a catastrophe in production. Human negligence can lead to increased downtimes and production failure.



IDOS ensures you overcome these challenges both cost-efficiently and at business speed, to deliver consistent performance while assuring absolute reliability for your industrial requirement.

Leap ahead with IDOS next-gen PSA technology

With the **IDOS** Pressure Swing Adsorption (PSA) oxygen generator, you ensure that your factory is using the latest clean technology, where the raw material is ambient air. The cost efficiency of a PSA oxygen generator coupled with the clean-tech, and easy-to-install-and-implement solution, make it **the preferred choice** over high-pressure cylinders and liquid oxygen tanks.

Capitalise on the advantages of an on-site PSA oxygen generator

Non-stop availability or 24 hours a day, 7 days a week

Independent supply of industrial oxygen

No safety hazards, hassle-free operation

Easy integration with existing installations

Reduced operational cost, no rental charges, logistical expenses and bulk user evaporation losses.

The only raw material **IDOS** uses is atmospheric air

The Industrial oxygen generator consists of two adsorbers filled with zeolite molecular sieves.

Pressurized air is passed through one of these adsorbers, zeolite having a structure to attract nitrogen, adsorbs nitrogen from air and oxygen is allowed to pass through to the storage tank and is produced as the final product.

As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of industrial oxygen.

This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of oxygen produced is continuously monitored by an oxygen analyzer.

IDOS is perfectly suited for a wide range of applications

- Fish Farming
- Wastewater Treatment
- Paper and Pulp Industry
- Glass Manufacturing
- Welding and Metal Fabrication
- Pharmaceutical and Biotech applications
- Plasma Cutting
- Laser Cutting
- Ceramic Industry
- Waste Incineration
- Refineries

Certification and compliances

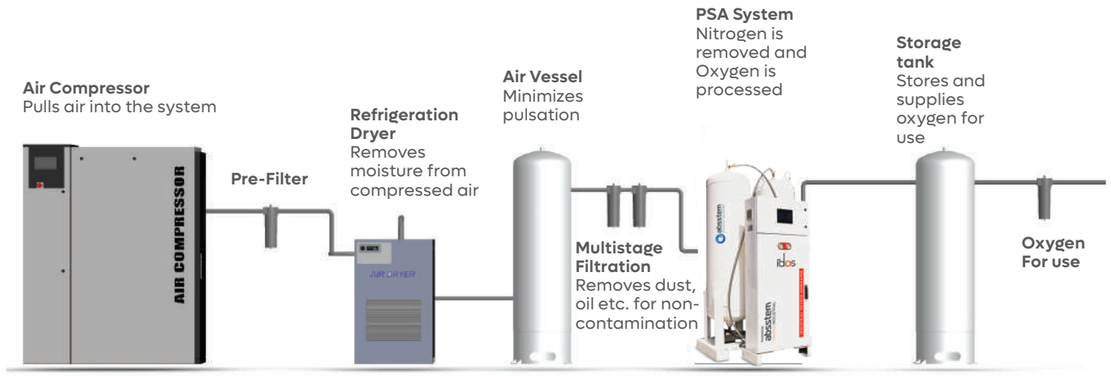
Our Quality Management Systems (QMS) is in full compliance with **ISO 9001:2015**

Our effective environmental management system (EMS) is in full compliance with **ISO 14001:2015**

Our occupational health and safety management systems are in full compliance with **ISO 45001:2018**

IDOS delivers the best-in-industry performance:

efficient, reliable, and consistent industrial oxygen generation.



Get the best-in-class Air Factor resulting in the lowest Compressed Air requirement. Additionally, with IDOS you benefit from:

- Best-in-class ROI
- Lowered Operational Cost
- Quiet and silent operation
- Lowered carbon footprint
- Flexibility to upgrade to high purity onsite

The Complete Scope of Supply comes with

- Zirconia/Electro chemical sensor for real-time purity
- Oxygen flow meter (Optional)
- Outlet pressure reducing valve (Optional)
- Oxygen pressure dewpoint sensor (Optional)

Boost capability with IDOS advanced features.

Reliable high flow capacity

The wide product range and gas flows exceeding 3,000 Nm³/h make these generators ideal for a variety of demanding applications.

Exceptional performance in tough weather conditions

- 24x7 Continuous production
- Built to last extreme weather conditions and tough terrains
- Show longevity in a dust or moisture-laden environments
- Ergonomic design, robust structure and advanced HMI
- Production Breakdown risks and downtimes are nullified

Achieve best-in-Class ROI

- Very Low operating expenses
- No additional costs such as order processing, refills and delivery charges
- Limited maintenance costs

Proven use-cases, it's always Plug and Plug

- Fully automated and monitored including the Oxygen sensor
- The only inputs required are air and electricity
- Get the desired purity your application requires

Always ahead.

Industry-leading technical features.

Highest Quality Zeolite Molecular Sieve (ZMS)

- Very High Density due to packed-bed Technology
- Protected by the process safety interlocks.
- 10 years warranty on ZMS*

Self-protective monitoring of the feed air quality parameters

- Temperature
- Pressure
- Pressure dewpoint
- Automatic feed air blow-off in case of contamination

Advanced HD Touch Screen Display

- Fully automatic, user friendly
- Remote monitoring equipped
- Advanced GUI with data monitoring and storage
- Oxygen purity readings in real-time
- Triggered alerts and alarms on any abnormal functionality
- Data storage with external storage capability for audit purposes
  - Can be remotely activated
  - RS232, RS422, USB and Ethernet ports supported

Multistage filtration system

- Filters remove oil and dust particles
- Additional filters drain moisture using an ADV (Automatic Drain Valve)

Automatic start-up

- Eliminates the risk of overflow and ZMS damage

Premium energy efficiency

- Lowest Air factor (Air-to-Oxygen ratio) in the Industry - 1.8 (95% O_2)

The ultimate energy saver

- Self-regulation and stable purity
- Stand-by mode in case no Oxygen is consumed
- Automatically regulates the requested oxygen pressure and purity



** When the plant is under CMC*

Technical Specifications

Relative humidity	Max 95%
Oxygen Flow:	2.5 to 3000 NM^3/hr
Working temperature	- 10°C to 55°C
Power Supply	230 VAC, 50/60 Hz
Inlet air pressure	Min 7.0 barg
O_2 Delivery Pressure	Up to 5.5 barg
Oxygen Dew Point	- 50°C – (Higher dew point available on demand)
ISO 9001:2015	Our Quality Management Systems is compliant

Technical Highlights

Lowest specific power in the industry	Inlet air as per **ISO 8573.1:2010 - Class 1.4.1**	**Purity testing done** by NABL APPROVED lab
Pressure up to **350 kg/cm²g** is available on-demand through oxygen compressors	**Sound Level of 65 dB** from a distance of 1 meter	**USB Storage** to store parameters values
CE-certified electronics with 24 VDC Operating Voltage	All tanks are fabricated per **ASME standards**	**Corrosion-resistant** internal environment for all tanks

Choose the model that best fits your requirements.

MODELS	PURITY (90-96%)	
	NM³/HR	LPM
IDOS 01	1.6	26
IDOS 02	2	33
IDOS 03	3	50
IDOS 04	4	67
IDOS 05	5.5	92
IDOS 06	6.5	108
IDOS 07	7.5	126
IDOS 09	9	150
IDOS 12.5	12.5	208
IDOS 15	15	250
IDOS 18	18	300
IDOS 22	22	367
IDOS 25	25	417
IDOS 30	30	500
IDOS 40	40	667
IDOS 45	45	750
IDOS 50	50	834
IDOS 65	65	1084
IDOS 75	75	1250
IDOS 90	90	1500
IDOS 110	110	1834
IDOS 130	130	2167
IDOS 150	150	2501
IDOS 200	200	3334
IDOS 250	250	4168
IDOS X	AS PER REQUIREMENT	

IDOS offers diverse Packaged Solutions

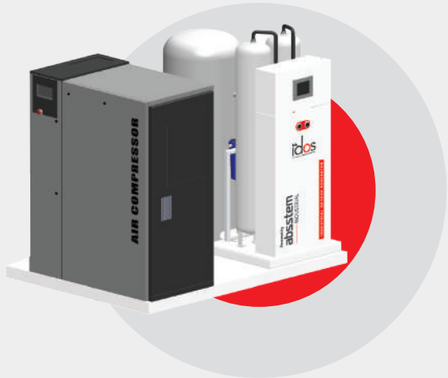


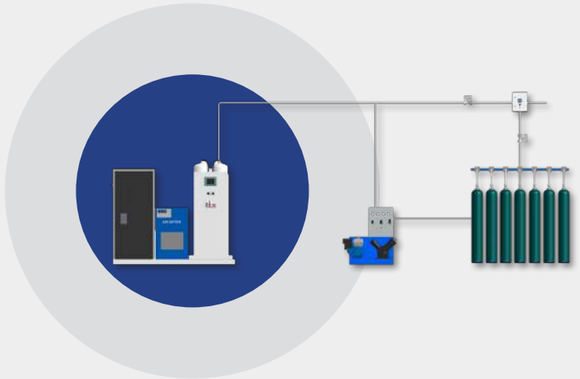
PSA Oxygen Generator

This is a plug-and-play system that is connected to your air compressor and storage tanks.

Skid - Package System

This system comes with an air compressor, multistage filtration system and a de-greased storage vessel. This is a complete package with all instruments required for oxygen generation.





High-pressure Oxygen Cylinder Filling System

Idos with an Oxygen Booster compressor for filling of cylinders.
Pressure – Up to 150 barg
Filling range – 8-1000 cylinders/day

Containerized Mobile PSA Oxygen Generator

Skid-mounted mobile systems with complete piping and interlocking support. They can be transported to any location and designed according to climate conditions and environmental factors. Cylinder filling options are available with noise control.



A world-standard plan, with pan-India service.

Absstem Shield

Premium Service across India

Absstem is a truly national organization with support available in more than 25 states. As a result, **our field service engineers** are never far away. We pride ourselves on the swift assistance that keeps your idos oxygen system performing reliably and efficiently.



Our Service Plans keep your IDOS oxygen systems in excellent shape. We have CMC options that could look after your installations for as long as 10 years.

10 year CMC Plans
We have solutions that look after your machines for as long as 10 years

Available 24/7. We keep spare parts in stock so your machine is up and running as quickly as possible

Quickest Turn Around Time in the industry – Our shield engineer is assigned to you within 10 minutes

For specific upgrades, add-ons and customised requirements, **please speak to us.**

Robust countrywide presence

200+ robust footprint nationwide

Present in over 31 states and Union Territories

Our wide service network covers expand to all states

absstem
REINVENTING PERFORMANCE

About Absstem

Innovation is as essential to our team as oxygen is to life. So how do we constantly innovate? By disrupting the existing category codes. We invest not only in gas separation of the purest grade but also maintain high standards in our after sales service. Pioneering the PSA technology in India, we deal with complete range of gas equipment plants. Keeping the pioneer spirit alive and well, we have become the market leaders in gas separation.

When it comes to the industrial sector, the requirements are always specific and the applications always demand a specific focus. The solutions to their problems are met with improved outputs and better efficiency - making it a success story for all parties involved. Absstem's research and development team took this into account and conceptualized a sustainable engineering solution for their oxygen needs which took care of their problem statements such as operator needs, consistent flow and continuous purity levels.

Best known for our presence in the on-site oxygen generation in Indian industrial sector we are also recognised by Dept. of Industrial Policy & Promotion for innovation in the industrial sector and registered under the Ministry of Micro, Small and Medium Enterprises, Government of India.

Corporate Address:
HQ - Plot No. 591, Sector-8, Phase-IV, IMT Manesar, Gurugram, Haryana

Work Address:
Plot No. 222 & 223 Industrial Area, Puth Khurd, New Delhi, Delhi - 110 039

Find us on info@absstem.com | Sales/Support Number - 1800-3010-3394, 92891 32903

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